



Regular graphs with positive Ollivier–Ricci curvature

Moritz Hehl¹

Received: 5 November 2024 / Revised: 3 January 2026 / Accepted: 9 January 2026 /
Published online: 11 February 2026
© The Author(s) 2026

Abstract

We study the Ollivier–Ricci curvature and its modification introduced by Lin, Lu, and Yau on regular graphs. We derive precise formulas for the Ollivier–Ricci curvature and the Lin–Lu–Yau curvature on regular graphs, expressed in terms of graph parameters and an optimal assignment between neighborhoods. Using these precise formulas, we provide an efficient implementation for computing the Ollivier–Ricci curvature and the Lin–Lu–Yau curvature on regular graphs, enabling us to explore the distribution of regular graphs with positive Lin–Lu–Yau curvature. Our main contribution is establishing a lower bound on the degree of a regular graph that guarantees positive Lin–Lu–Yau curvature and non-negative Ollivier–Ricci curvature, along with an analysis of the sharpness of this bound. Additionally, we study the maximal number of vertices that an d -regular graph with positive Lin–Lu–Yau curvature can have.

Mathematics Subject Classification 53C21 · 05C81 · 05C30 · 68R10

1 Introduction and statement of results

Since Gauss and Riemann introduced the geometric notion of curvature over 150 years ago, it played a central role in differential geometry. Among the various types of curvature, Ricci curvature is of particular importance, serving as a fundamental tool in the study of Riemannian manifolds. Given the important role of Ricci curvature in differential geometry, there is a natural interest in extending this concept to metric spaces more general than Riemannian manifolds. This has led to the proposal of various generalized notions of curvature for non-smooth or discrete structures (see, e.g., [1–4]). Each approach captures a different aspect of the Riemannian notion, and each is thus suited for a different purpose.

Von Renesse and Sturm [5] established a connection between Ricci curvature and optimal transport on smooth Riemannian manifolds. Building on their results, Ollivier [6] introduced a discrete notion of Ricci curvature on metric spaces equipped with Markov chains or a measure. His approach leverages optimal transport theory, as his

✉ Moritz Hehl
moritz.hehl@uni-leipzig.de

¹ Institute of Mathematics, Universität Leipzig, 04109 Leipzig, Germany

definition of curvature is based on the Wasserstein distance. For a precise definition a detailed introduction is required, which is provided in Sect. 2.1.

In the context of locally finite graphs, Ollivier's notion of Ricci curvature has recently received considerable attention. In this setting, a modification of Ollivier's notion was introduced by Lin, Lu, and Yau in [7], the *Lin–Lu–Yau curvature*. The first contribution of this work is the derivation of a precise formula for the Lin–Lu–Yau curvature on regular graphs using elementary means.

Theorem 1.1 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Then the Lin–Lu–Yau curvature*

$$\kappa(x, y) = \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in S_1(x) \setminus B_1(y)} d(z, \phi(z)) \right),$$

where $\mathcal{A}_{xy}$ denotes the set of assignments between $S_1(x) \setminus B_1(y)$ and $S_1(y) \setminus B_1(x)$.

Bonini et al. [8, Theorem 4.6] derived an exact formula for the Lin–Lu–Yau curvature on strongly regular graphs in terms of graph parameters and the size of a maximal matching in the core neighborhood. In [9], Münch et al. provided an exact formula for a modified curvature notion on combinatorial graphs. To the best of my knowledge, precise expressions for the Lin–Lu–Yau curvature on arbitrary regular graphs have not yet been established.

We then use the exact formula for the Lin–Lu–Yau curvature to provide an efficient implementation for computing the Lin–Lu–Yau curvature on regular graphs. This allows us to investigate the distribution of regular graphs with positive Lin–Lu–Yau curvature, leading to the main result of this work: The establishment of a lower bound on the degree of a regular graph that ensures positive Lin–Lu–Yau curvature.

Theorem 1.2 *Let $G = (V, E)$ be a finite graph. If G is d -regular with*

$$d > \frac{2|V|}{3} - 2,$$

then $\kappa(x, y) > 0$ for every edge $x \sim y$.

A similar result holds true for the Ollivier–Ricci curvature.

Theorem 1.3 *Let $G = (V, E)$ be a finite graph. If G is d -regular with*

$$d > \frac{2|V|}{3} - 2,$$

then $\kappa_0(x, y) \geq 0$ for every edge $x \sim y$.

After deriving this lower bound on the degree of regular graphs that guarantees positive Lin–Lu–Yau curvature and non-negative Ollivier–Ricci curvature, we proceed to discuss the sharpness of the bound.

Theorem 1.4 *Let $n \in \mathbb{N}$ such that*

$$d = \frac{2n - 8}{3} \in 2\mathbb{N}$$

and $d \geq 12$. Then, there exists a d -regular graph on n vertices such that $\kappa(x, y) = 0$ and $\kappa_0(x, y) < 0$ for an edge $x \sim y$.

Finally, we study the maximal number of vertices that a d -regular graph G with positive Lin–Lu–Yau curvature can have,

$$\mathcal{M}_d = \max_{\substack{G \text{ } d\text{-regular,} \\ \text{Ric}(G) > 0}} |V(G)|.$$

We prove that $\mathcal{M}_d$ is finite for every degree d and we obtain precise values for d equal to two and three.

Theorem 1.5 *Let $G = (V, E)$ be a d -regular graph, satisfying $\text{Ric}(G) > 0$. If $d = 2$, then $|V| \leq 5$. If $d = 3$, then $|V| \leq 10$. Consequently, we have $\mathcal{M}_2 = 5$ and $\mathcal{M}_3 = 10$.*

We conclude this introduction by providing an outline of the remainder of the paper. In Sect. 2 we review the relevant concepts of graph theory and optimal transport theory, and introduce the Ollivier–Ricci curvature, as well as its modification by Lin, Lu, and Yau. In Sect. 3 we derive exact formulas for the different curvature notions. In Sect. 4 we present our efficient implementation for the computation of the Lin–Lu–Yau curvature and explore the distribution of regular graphs with positive Lin–Lu–Yau curvature. In Sect. 5, we establish our main result. Namely, we derive a lower bound on the degree of a regular graph that ensures positive Lin–Lu–Yau curvature. Finally, in Sect. 6, we investigate the maximal number of vertices that a d -regular graph with positive Lin–Lu–Yau curvature can have and present some open questions.

2 Definitions and notations

We begin by reviewing some fundamental concepts of graph theory and optimal transport theory. We then introduce Ollivier’s discrete notion of Ricci curvature on graphs, as well as its modification by Lin, Lu, and Yau.

2.1 Graph theory

A *simple graph* $G = (V, E)$ is an unweighted, undirected graph that contains no multiple edges or self-loops. For two vertices $x, y \in V$ we denote the existence of an edge between x and y by $x \sim y$. For any two vertices $x, y \in V$, the *shortest-path distance* $d(x, y)$ is the number of edges in a shortest path connecting x and y . If no such path exists, $d(x, y)$ is defined to be infinity. For two nonempty sets $A, B \subseteq V$, we define

$$\text{dist}(A, B) = \inf \{d(x, y) : x \in A, y \in B\}.$$

The *diameter* of G is denoted by $\text{diam}(G) = \max_{x,y \in V} d(x, y)$. For $x \in V$ and $r \in \mathbb{N}$ we define the r -sphere centered at x as $S_r(x) = \{y \in V : d(x, y) = r\}$ and the r -ball centered at x as $B_r(x) = \{y \in V : d(x, y) \leq r\}$. For an edge $x \sim y$, we denote by $\Delta(x, y) = S_1(x) \cap S_1(y)$ the set of common neighbors of x and y .

The *degree* of a vertex $x \in V$ is denoted by $d_x = |S_1(x)|$. The graph G is called *locally finite* if every vertex has finite degree. The graph is said to be d -regular if every vertex has the same degree d .

The Moore bound provides an upper bound on the number of vertices a regular graph with given diameter can have.

Theorem 2.1 (Moore bound) *Let $G = (V, E)$ be a d -regular graph with diameter k . Then*

$$|V| \leq 1 + d \sum_{i=0}^{k-1} (d-1)^i.$$

A simple graph $G = (V, E)$ is called *bipartite*, if there is a partition of V into two disjoint sets $V = A \cup B$, such that every edge connects a vertex in A to a vertex in B . If the partition $V = A \cup B$ is specified, we denote the bipartite graph by $G = (A, B, E)$.

In what follows, the famous *Hall's marriage theorem* [10] will be of importance. Before stating the theorem, we must first introduce the concept of a perfect matching. For a simple graph $G = (V, E)$, we call a subset of edges $S \subseteq E$ a *matching* of G if no vertex in V is incident to more than one edge in S . For a bipartite graph $G = (A, B, E)$, we call a matching S a *perfect matching of A into B* if every vertex in A is incident to exactly one edge in S .

Theorem 2.2 (Hall's marriage theorem) *Let $G = (A, B, E)$ be a bipartite graph. Then there exists a perfect matching of A into B if and only if $|N(X)| \geq |X|$ for every $X \subseteq A$, where $N(X) = \bigcup_{x \in X} S_1(x)$.*

As a consequence of Hall's marriage theorem, we obtain the following Corollary.

Corollary 2.3 *Let $G = (A, B, E)$ be a bipartite graph. If G is regular, then there exists a perfect matching of A into B .*

In what follows, $G = (V, E)$ is always assumed to be a locally finite, connected, and simple graph.

2.2 Optimal transport theory

A probability measure on the vertex set V is a function $\mu : V \rightarrow [0, 1]$ that satisfies $\sum_{x \in V} \mu(x) = 1$. We denote by $\mathcal{P}(V)$ the set of probability measures on V . The Wasserstein distance, a metric on the set $\mathcal{P}(V)$, is a fundamental concept in optimal transport theory.

Definition 2.4 (*Wasserstein distance*) Let $G = (V, E)$ be a locally finite graph. Let μ_1, μ_2 be two probability measures on V . The *Wasserstein distance* between μ_1 and μ_2 is defined as

$$W_1(\mu_1, \mu_2) = \inf_{\pi \in \Pi(\mu_1, \mu_2)} \sum_{x \in V} \sum_{y \in V} d(x, y) \pi(x, y), \quad (2.1)$$

where

$$\Pi(\mu_1, \mu_2) = \left\{ \pi : V \times V \rightarrow [0, 1] : \sum_{y \in V} \pi(x, y) = \mu_1(x), \sum_{x \in V} \pi(x, y) = \mu_2(y) \right\}.$$

Intuitively, imagine two distributions given by μ_1 and μ_2 as piles of earth. The Wasserstein distance measures the minimal effort required to transform one pile of earth into another. We call $\pi \in \Pi(\mu_1, \mu_2)$ a *transport plan* and if the infimum in 2.1 is attained, we call π an *optimal transport plan* transporting μ_1 to μ_2 .

The following Lemma allows us to impose some assumptions on an optimal transport plan. It states that it is not necessary to move any mass that is shared between the two probability measures.

Lemma 2.5 *Let $G = (V, E)$ be a locally finite graph. Let μ_1, μ_2 be two probability measures on V . Then there exists an optimal transport plan π transporting μ_1 to μ_2 satisfying*

$$\pi(x, x) = \min\{\mu_1(x), \mu_2(x)\}$$

for all $x \in V$.

The proof is an immediate consequence of the triangle inequality. For more details, we refer to [12, Lemma 4.1]

In what follows, probability measures under which all points in their support carry equal mass will be of importance, i.e.,

$$\mu_1 = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}, \quad \mu_2 = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}.$$

Here, δ_x denotes the Dirac measure centered on some fixed vertex $x \in V$:

$$\delta_x(y) = \begin{cases} 1, & \text{if } x = y; \\ 0, & \text{otherwise.} \end{cases}$$

In this case, any transport plan $\pi \in \Pi(\mu_1, \mu_2)$ can be represented as a bistochastic $n \times n$ matrix $\pi = (\pi_{ij})_{i,j=1,\dots,n}$. Here bistochasticity means that $\pi_{ij} \geq 0$ and that

$$\sum_{k=1}^n \pi_{ik} = \sum_{k=1}^n \pi_{kj} = 1$$

for any $i, j \in \{1, \dots, n\}$. We denote the set of all bistochastic $n \times n$ matrices by $\mathcal{B}_n$. Thus, the Wasserstein distance reduces to

$$W_1(\mu_1, \mu_2) = \inf_{\pi \in \mathcal{B}_n} \frac{1}{n} \sum_{i,j=1}^n \pi_{ij} d(x_i, y_j).$$

Therefore, calculating the Wasserstein distance is a linear optimization problem on the bounded and convex set $\mathcal{B}_n$. *Choquet's minimization theorem* states that some extremal points of $\mathcal{B}_n$ are optimal. *Birkhoff's theorem* states that these extremal points are permutation matrices, i.e.,

$$\pi_{ij} = \delta_{\sigma(i),j} = \begin{cases} 1, & \text{if } \sigma(i) = j; \\ 0, & \text{otherwise.} \end{cases}$$

for some permutation σ of $\{1, \dots, n\}$. Here δ_{ij} is the Kronecker symbol. We denote by S_n the set of all permutations of $\{1, \dots, n\}$. Thus, we obtain the following Lemma [11].

Lemma 2.6 *Let $G = (V, E)$ be a locally finite graph. Let $\mu_1 = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ and $\mu_2 = \frac{1}{n} \sum_{i=1}^n \delta_{y_i}$, where $\{x_1, \dots, x_n\} \subset V$ and $\{y_1, \dots, y_n\} \subset V$. Then the Wasserstein distance between μ_1 and μ_2 reduces to*

$$W_1(\mu_1, \mu_2) = \inf_{\sigma \in S_n} \frac{1}{n} \sum_{i=1}^n d(x_i, y_{\sigma(i)}).$$

In other words, there exists an optimal transport plan that corresponds to a one-to-one assignment between the source vertices $x_1, \dots, x_n$ and the target vertices $y_1, \dots, y_n$. We refer to such an optimal transport plan as an *optimal assignment*. In other words, the Kantorovich problem coincides with the Monge problem in this special case.

2.3 Ollivier–Ricci curvature and its modification

To introduce Ollivier's notion of Ricci curvature on graphs, we define the probability measures μ_x^α for $x \in V$ and $\alpha \in [0, 1]$ by

$$\mu_x^\alpha(y) = \begin{cases} \alpha, & \text{if } y = x; \\ \frac{1-\alpha}{d_x}, & \text{if } y \sim x; \\ 0, & \text{otherwise.} \end{cases}$$

Then, the Ollivier–Ricci curvature is defined as follows.

Definition 2.7 (*Ollivier–Ricci curvature*). Let $G = (V, E)$ be a locally finite graph. We define the α -Ollivier–Ricci curvature of an edge $x \sim y$ by

$$\kappa_\alpha^G(x, y) = 1 - W_1^G(\mu_x^\alpha, \mu_y^\alpha).$$

The parameter α is called the *idleness*.

Ollivier considered idleness parameters $\alpha = 0$ and $\alpha = \frac{1}{2}$. In [12], the authors study the Ollivier–Ricci curvature as a function of the idleness parameter. To this end, they introduced the *Ollivier–Ricci idleness function* $\alpha \rightarrow \kappa_\alpha(x, y)$. It was first shown by Lin, Lu, and Yau in [7], that the idleness function is concave. Using that $\kappa_1(x, y) = 0$, this implies that the function $h(\alpha) = \frac{\kappa_\alpha(x, y)}{1 - \alpha}$ is increasing over the interval $[0, 1)$. They further showed that $h(\alpha)$ is bounded and thus, the limit $\lim_{\alpha \rightarrow 1} h(\alpha)$ exists. Lin, Lu, and Yau used this result to introduce a modification of the Ollivier–Ricci curvature.

Definition 2.8 (*Lin–Lu–Yau curvature*). Let $G = (V, E)$ be a locally finite graph. The *Lin–Lu–Yau curvature* of an edge $x \sim y$ is defined as

$$\kappa(x, y) = \lim_{\alpha \rightarrow 1} \frac{\kappa_\alpha(x, y)}{1 - \alpha}.$$

Remark 2.9 Observe that $\kappa_1(x, y) = 0$ for any edge $x \sim y$. Thus, $\kappa(x, y)$ is the negative of the derivative of $\kappa_\alpha(x, y)$ with respect to the idleness parameter in $\alpha = 1$.

In what follows, we write $\text{Ric}(G) > 0$ if $\kappa(x, y) > 0$ for all edges $x \sim y$ in G .

Bourne et al. [12] showed that the idleness function is piecewise linear with at most 3 linear parts. They also derived the length of the last linear part.

Theorem 2.10 [12, Theorem 4.4]. *Let $G = (V, E)$ be a locally finite graph and let $x, y \in V$ with $x \sim y$ and $d_x \geq d_y$. Then $\alpha \rightarrow \kappa_\alpha(x, y)$ is linear over $\left[\frac{1}{d_x+1}, 1\right]$.*

Thus, we obtain the following relation between the α -Ollivier–Ricci curvature and its modification by Lin, Lu, and Yau as an immediate consequence of the mean value theorem.

Theorem 2.11 *Let $G = (V, E)$ be a locally finite graph and let $x, y \in V$ with $x \sim y$ and $d_x \geq d_y$. Then*

$$\kappa_\alpha(x, y) = (1 - \alpha)\kappa(x, y)$$

for $\alpha \in \left[\frac{1}{d_x+1}, 1\right]$.

Hence, the Lin–Lu–Yau curvature coincides up to a scaling factor with the α -Ollivier–Ricci curvature for large values of α .

3 Exact formulas for the Ollivier–Ricci curvature

In this section, we shall derive precise formulas for the Lin–Lu–Yau curvature and the α -Ollivier–Ricci curvature of an edge $x \sim y$ with $d_x = d_y$.

3.1 Decomposing the neighborhood

To derive our precise formulas, we decompose the 1-spheres centered at the vertices x and y into disjoint sets. We denote by $\Delta(x, y)$ the set of common neighbors of both x and y , namely,

$$\Delta(x, y) = S_1(x) \cap S_1(y).$$

Furthermore, we define $R_x(x, y) = S_1(x) \setminus B_1(y)$ and $R_y(x, y) = S_1(y) \setminus B_1(x)$, giving rise to the following disjoint decompositions of $S_1(x)$ and $S_1(y)$:

$$S_1(x) = \{y\} \cup \Delta(x, y) \cup R_x(x, y) \quad \text{and} \quad S_1(y) = \{x\} \cup \Delta(x, y) \cup R_y(x, y).$$

Using these decompositions, we show that the computation of the Wasserstein distance reduces to an assignment problem between $R_x(x, y)$ and $R_y(x, y)$. Therefore, we introduce the concept of optimal assignments.

Definition 3.1 (*Optimal assignment*). Let $G = (V, E)$ be a locally finite graph. Let $U, W \subseteq V$ with $|U| = |W| < \infty$. We call a bijection $\phi : U \rightarrow W$ an *assignment* between U and W . Denote by $\mathcal{A}_{UW}$ the set of all assignments between U and W . We call $\phi \in \mathcal{A}_{UW}$ an *optimal assignment* between U and W if

$$\sum_{z \in U} d(z, \phi(z)) = \inf_{\psi \in \mathcal{A}_{UW}} \sum_{z \in U} d(z, \psi(z)).$$

Denote the set of all optimal assignments between U and W by $\mathcal{O}_{UW}$.

Note that for an edge $x \sim y$ with $d_x = d_y$, we have $|R_x(x, y)| = |R_y(x, y)|$. For the sake of simplicity, we will denote by $\mathcal{A}_{xy}$ and $\mathcal{O}_{xy}$ the sets $\mathcal{A}_{R_x(x, y)R_y(x, y)}$ and $\mathcal{O}_{R_x(x, y)R_y(x, y)}$, respectively.

3.2 Ollivier–Ricci curvature on the last linear part

To derive our formula for the Lin–Lu–Yau curvature, we begin by deriving an expression for the Ollivier–Ricci curvature on the last linear part. We then utilize the fact that on this segment, the Lin–Lu–Yau curvature coincides with the Ollivier–Ricci curvature, differing only by a scaling factor.

We first need the following Lemma about some of the assumptions we can impose on an optimal transport plan transporting μ_x^α to μ_y^α .

Lemma 3.2 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Furthermore, let $\alpha \in \left[\frac{1}{d+1}, 1\right]$. Then there exists an optimal transport plan π transporting μ_x^α to μ_y^α with the following properties:*

- (i) $\forall i \in \Delta(x, y) : \pi(i, i) = \mu_x^\alpha(i)$;
- (ii) $\pi(x, x) = \pi(y, y) = \mu_y^\alpha(x)$;
- (iii) $\pi(x, y) = \mu_y^\alpha(y) - \mu_y^\alpha(x)$.

Remark 3.3 Observe that $\mu_y^\alpha(y) - \mu_y^\alpha(x) \geq 0$ if and only if $\alpha \geq \frac{1}{d+1}$. Hence, the subsequent discussion pertains only to the last linear part of κ_α .

Proof Properties (i) and (ii) are immediate consequence of Lemma 2.5.

For the proof of (iii), let π be an optimal transport plan transporting μ_x^α to μ_y^α while satisfying properties (i) and (ii). Since π satisfies property (ii), we must have

$$\begin{aligned}\pi(x, y) &= \mu_y^\alpha(y) - \sum_{j \in V \setminus \{x\}} \pi(j, y) \\ &\leq \mu_y^\alpha(y) - \pi(y, y) \\ &= \mu_y^\alpha(y) - \mu_y^\alpha(x).\end{aligned}$$

Now, assume that $\pi(x, y) < \mu_y^\alpha(y) - \mu_y^\alpha(x)$. Thus, the sets

$$I = \{z \in S_1(x) \setminus \{y\} : \pi(z, y) > 0\} \quad \text{and} \quad O = \{z \in S_1(y) \setminus \{x\} : \pi(x, z) > 0\}$$

are non-empty. Due to property (i), we must have $I \cap O = \emptyset$. Since π satisfies property (ii), we obtain:

$$\begin{aligned}\sum_{z \in I} \pi(z, y) &= \mu_y^\alpha(y) - \pi(y, y) - \pi(x, y) \\ &= \mu_x^\alpha(x) - \pi(x, x) - \pi(x, y) \\ &= \sum_{z \in O} \pi(x, z).\end{aligned}$$

In other words, the mass transported from I to y equals the mass transported from x to O under π . We denote this mass by m , namely $m = \sum_{z \in I} \pi(z, y)$.

We now obtain a new transport plan π' from modifying π as follows:

- Set $\pi'(x, y) = \mu_y^\alpha(y) - \mu_y^\alpha(x)$, $\pi'(x, z) = 0 \forall z \in O$ and $\pi'(z, y) = 0 \forall z \in I$
- Move the mass m directly from I to O .

See Fig. 1 for an illustration of this modification.

Due to property (i), we must have $O \cap \Delta(x, y) = I \cap \Delta(x, y) = \emptyset$. Therefore, $d(i, y) = 2$ for any $i \in I$ and $d(x, j) = 2$ for any $j \in O$. Thus, π moves mass

- m at distance two from I to y ,
- m at distance two from x to O ,
- $\pi(x, y) = \mu_y^\alpha(y) - \mu_y^\alpha(x) - m$ at distance one from x to y ,

resulting in a cost of:

$$3 \cdot m + \mu_y^\alpha(y) - \mu_y^\alpha(x).$$

On the other hand, observe that $d(i, j) \leq 3$ for any $i \in I$ and $j \in O$. Thus, π' moves the mass

- $\mu_y^\alpha(y) - \mu_y^\alpha(x)$ at distance one from x to y ,
- m from I to O at distance less than or equal to three,

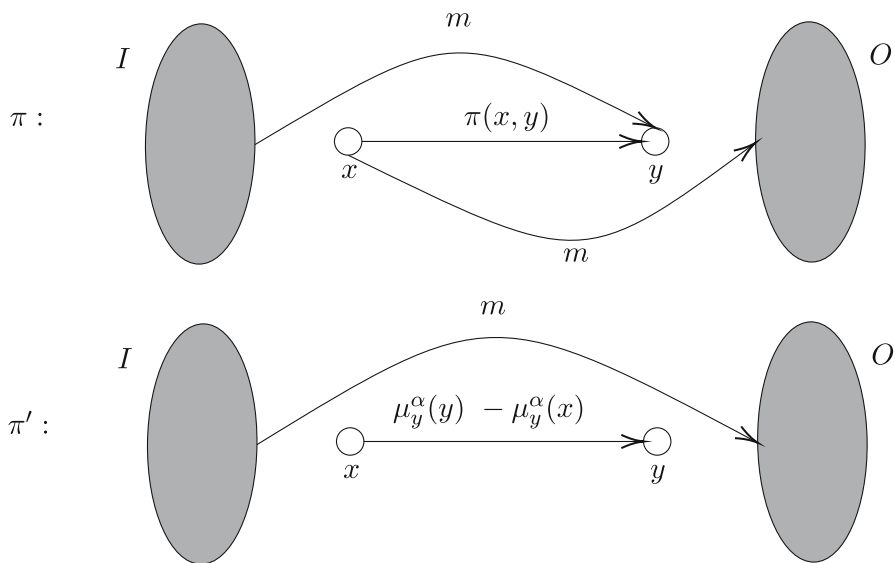


Fig. 1 Illustration of the optimal transport plan π and its modification π' for the proof of Lemma 3.2

resulting in a cost less than or equal to:

$$3 \cdot m + \mu_y^\alpha(y) - \mu_y^\alpha(x).$$

This is exactly the cost of moving this part of the mass under the optimal transport plan π . As π' coincides with π everywhere else, π' is an optimal transport plan as well. Finally, observe that by construction,

$$\pi'(j, j) = \pi(j, j) \quad \forall j \in \Delta(x, y), \quad \pi'(x, x) = \pi(x, x) \quad \text{and} \quad \pi'(y, y) = \pi(y, y).$$

Hence, we have constructed an optimal transport plan satisfying properties (i)–(iii). $\square$

We are now ready to derive a precise expression for the Ollivier–Ricci curvature on the last linear part.

Theorem 3.4 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Furthermore, let $\alpha \in \left[\frac{1}{d+1}, 1\right]$. Then the α -Ollivier–Ricci curvature*

$$\kappa_\alpha(x, y) = \frac{1 - \alpha}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right).$$

Remark 3.5 This result can be derived from the linearity of the idleness function together with properties (i), (ii), and (iii) in the proof of Theorem 1.1 in [19]. For the reader's convenience, we provide a full proof here.

Proof To demonstrate this, we proceed with a proof by cases.

Case 1: $R_x(x, y) = \emptyset$. In this case, $\mathcal{A}_{xy}$ only contains the empty function. Thus,

$$\frac{1-\alpha}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right) = \frac{1-\alpha}{d} (d + 1).$$

On the other hand, the transport plan π transporting μ_x^α to μ_y^α , defined by

$$\pi(i, j) = \begin{cases} \frac{1-\alpha}{d}, & \text{if } i = j \in \{x, y\} \cup \Delta(x, y); \\ \alpha - \frac{1-\alpha}{d}, & \text{if } i = x \text{ and } j = y; \\ 0, & \text{otherwise.} \end{cases}$$

is optimal, according to Lemma 3.2. Therefore,

$$W_1(\mu_x^\alpha, \mu_y^\alpha) = \sum_{i \in B_1(x)} \sum_{j \in B_1(y)} d(i, j) \pi(i, j) = \alpha - \frac{1-\alpha}{d},$$

and hence,

$$\kappa_\alpha(x, y) = \frac{1-\alpha}{d} (d + 1).$$

Case 2: $R_x(x, y) \neq \emptyset$. Let $\phi \in \mathcal{O}_{xy}$ be an optimal assignment between $R_x(x, y)$ and $R_y(x, y)$. Define the transport plan π transporting μ_x^α to μ_y^α by

$$\pi(i, j) = \begin{cases} \frac{1-\alpha}{d}, & \text{if } i = j \in \{x, y\} \cup \Delta(x, y); \\ \frac{1-\alpha}{d}, & \text{if } i \in R_x(x, y) \text{ and } j = \phi(i); \\ \alpha - \frac{1-\alpha}{d}, & \text{if } i = x \text{ and } j = y; \\ 0, & \text{otherwise.} \end{cases}$$

We claim that this transport plan is optimal. Assume on the contrary that is is not optimal. Then let π' be an optimal transport plan transporting μ_x^α to μ_y^α , satisfying properties (i)–(iii) stated in Lemma 3.2. Then,

$$\sum_{i \in B_1(x)} \sum_{j \in B_1(y)} \pi'(i, j) d(i, j) = \alpha - \frac{1-\alpha}{d} + \sum_{i \in R_x(x, y)} \sum_{j \in R_x(x, y)} \pi'(i, j) d(i, j). \quad (3.1)$$

On the other hand, we obtain by construction of π

$$\sum_{i \in B_1(x)} \sum_{j \in B_1(y)} \pi(i, j) d(i, j) = \alpha - \frac{1-\alpha}{d} + \sum_{i \in R_x(x, y)} \sum_{j \in R_x(x, y)} \pi(i, j) d(i, j). \quad (3.2)$$

Due to the non-optimality of π , we must have:

$$\sum_{i \in B_1(x)} \sum_{j \in B_1(y)} \pi'(i, j) d(i, j) < \sum_{i \in B_1(x)} \sum_{j \in B_1(y)} \pi(i, j) d(i, j). \quad (3.3)$$

Combining Eqs. 3.1, 3.2 and 3.3 leads to

$$\sum_{i \in R_x(x, y)} \sum_{j \in R_x(x, y)} \pi'(i, j) d(i, j) < \sum_{i \in R_x(x, y)} \sum_{j \in R_x(x, y)} \pi(i, j) d(i, j). \quad (3.4)$$

Now, observe that $\pi|_{R_x(x, y) \times R_y(x, y)}$ and $\pi'|_{R_x(x, y) \times R_y(x, y)}$ are transport plans transporting $\mu_x^\alpha|_{R_x(x, y)}$ to $\mu_y^\alpha|_{R_y(x, y)}$. Furthermore, $\mu_x^\alpha|_{R_x(x, y)}$ and $\mu_y^\alpha|_{R_y(x, y)}$ are measures under which all points in their support carry equal mass $\frac{1-\alpha}{d}$.

Thus, we can apply Lemma 2.6 (after multiplication with a scaling factor) and conclude that $\pi|_{R_x(x, y) \times R_y(x, y)}$ is an optimal transport plan. This contradicts Eq. 3.4.

Therefore, our assumption was incorrect and π is an optimal transport plan. The rest of the proof is just a straightforward calculation. Namely,

$$\begin{aligned} W_1(\mu_x^\alpha, \mu_y^\alpha) &= \sum_{i \in B_1(x)} \sum_{j \in B_1(y)} d(i, j) \pi(i, j) \\ &= \alpha - \frac{1-\alpha}{d} + \frac{1-\alpha}{d} \sum_{z \in R_x(x, y)} d(z, \phi(z)), \end{aligned}$$

which leads to

$$\kappa_\alpha(x, y) = \frac{1-\alpha}{d} \left(d + 1 - \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right).$$

The fact that ϕ is an optimal assignment concludes the proof. $\square$

3.3 Lin–Lu–Yau curvature

According to Theorem 2.11, the identity

$$\kappa(x, y) = \frac{\kappa_\alpha(x, y)}{1-\alpha}$$

holds true on the last linear part of the idleness function. Therefore, our formula for the Lin–Lu–Yau curvature is an immediate consequence of Theorem 3.4.

Theorem 1.1 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Then the Lin–Lu–Yau curvature*

$$\kappa(x, y) = \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right). \quad (3.5)$$

Let us now state some consequence of our formula.

Corollary 3.6 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Then $\kappa(x, y) \in \mathbb{Z}/d$. Furthermore,*

$$-2 + \frac{4}{d} + 3 \frac{|\Delta(x, y)|}{d} \leq \kappa(x, y) \leq \frac{2 + |\Delta(x, y)|}{d}.$$

Proof The fact that $\kappa(x, y) \in \mathbb{Z}/d$ follows directly from formula 3.5.

Using that $d(z, \phi(z)) \geq 1$ for all $z \in R_x(x, y)$ and any assignment $\phi \in \mathcal{A}_{xy}$, we obtain

$$\begin{aligned} \kappa(x, y) &= \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right) \\ &\leq \frac{1}{d} \left(d + 1 - |R_x(x, y)| \right) \\ &= \frac{1}{d} \left(d + 1 - (d - 1 - |\Delta(x, y)|) \right) \\ &= \frac{2 + |\Delta(x, y)|}{d}. \end{aligned}$$

Similarly, using that $d(z, \phi(z)) \leq 3$, we obtain

$$\begin{aligned} \kappa(x, y) &= \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right) \\ &\geq \frac{1}{d} \left(d + 1 - 3|R_x(x, y)| \right) \\ &= \frac{1}{d} \left(-2d + 4 + 3|\Delta(x, y)| \right). \end{aligned}$$

□

3.4 Ollivier–Ricci curvature for vanishing idleness

This section aims to derive an exact formula for the α -Ollivier–Ricci curvature of an edge $x \sim y$ with $d_x = d_y$, in the case of vanishing idleness, i.e., $\alpha = 0$.

Recall that for $\alpha = 0$, μ_x^α simplifies to

$$\mu_x^\alpha(y) = \begin{cases} \frac{1}{d_x}, & \text{if } y \in S_1(x); \\ 0, & \text{otherwise.} \end{cases}$$

As per Lemma 2.6, the Kantorovich problem coincides with the Monge problem in this case. The following Lemma states some of the assumptions we can impose on an optimal assignment.

Lemma 3.7 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Then there exists an optimal assignment ϕ between $S_1(x)$ and $S_1(y)$, such that $\phi(i) = i$ for all $i \in \Delta(x, y)$.*

Using analogous reasoning as in the case of the Lin–Lu–Yau curvature, we obtain a precise expression for the 0-Ollivier–Ricci curvature.

Theorem 3.8 *Let $G = (V, E)$ be a locally finite graph. Let $x, y \in V$ be of equal degree d with $x \sim y$. Then*

$$\kappa_0(x, y) = \frac{1}{d} \left(d - \inf_{\phi} \sum_{z \in S_1(x) \setminus S_1(y)} d(z, \phi(z)) \right),$$

where the infimum is taken over all assignments ϕ between $S_1(x) \setminus S_1(y)$ and $S_1(y) \setminus S_1(x)$.

4 Distribution of positive Lin–Lu–Yau curvature in regular graphs

In this section, we wish to study the distribution of regular graphs G satisfying $\text{Ric}(G) > 0$. To do so, we are faced with the problem of generating exhaustive lists of all d -regular graphs with a given number of vertices, up to isomorphism. This is one of the oldest problems in the field of constructive combinatorics [13]. Already determining the number of d -regular graphs on n vertices, denoted by $|\mathcal{G}_{n,d}|$, proves to be an exceedingly difficult problem for which no closed formula is currently known. But there are some interesting asymptotic results. Wormald and McKay [14] conjectured that asymptotically

$$|\mathcal{G}_{n,d(n)}| \sim \sqrt{2} \binom{n-1}{d(n)}^n (\lambda^\lambda (1-\lambda)^{(1-\lambda)})^{\binom{n}{2}} e^{\frac{1}{4}},$$

where $\lambda = \frac{d(n)}{n-1}$ and $d(n)$ satisfying $1 \leq d(n) \leq n-2$ with $nd(n)$ always an even integer. A complete proof of this conjecture was only presented in 2017 [15].

For the asymptotic number of regular graphs up to isomorphism, the study of the automorphism groups of regular graphs is necessary. We refer the interested reader to the work by Wormald and McKay [16] for more details.

While no closed formula is available, algorithms designed to generate all d -regular graphs on n vertices have determined the number of such graphs, up to isomorphism, for small values of n . In the following, we use the algorithm *GENREG*, proposed by Meringer in [13]. Refer to Table 1 for the total number of connected d -regular graphs on n vertices.

Table 1 Number of connected d -regular graphs with given number of vertices n , up to isomorphism; empty fields indicate values that are not yet known [13]

d	3	4	5	6	7	8
n						
4	1	0	0	0	0	0
5	0	1	0	0	0	0
6	2	1	1	0	0	0
7	0	2	0	1	0	0
8	5	6	3	1	1	0
9	0	16	0	4	0	1
10	19	59	60	21	5	1
11	0	265	0	266	0	6
12	85	1,544	7,848	7,849	1,547	94
13	0	10,778	0	367,860	0	10,786
14	509	88,168	3,458,383	21,609,300	21,609,301	3,459,386
15	0	805,491	0	1,470,293,675	0	1,470,293,676
16	4060	8,037,418	2,585,136,675	113,314,233,808	733,351,105,934	733,351,105,935
17	0	86,221,634	0		0	
18	41301	985,870,522				
19	0	11,946,487,647	0		0	
20	510,489					

Given the rapid growth in the number of regular graphs with specified degree and vertex count, an efficient implementation for the computation of the Lin–Lu–Yau curvature is necessary to quantify the number of regular graphs G satisfying $Ric(G) > 0$. Current implementations do not provide sufficient efficiency for this task at scale.

Building on the precise formula derived in Theorem 1.1, we provide a new, efficient implementation for the computation of the Lin–Lu–Yau curvature on regular graphs. The source code is available in a public repository.¹ See Appendix A for a comparative analysis between our implementation and the existing implementations by Ni et al. [17], and by Cushing et al. [20]. On smaller graphs ($n \leq 20$) our implementation achieves a speedup of 20–50 times, depending on the exact values of d and n .

Using our efficient implementation for the computation of the Lin–Lu–Yau curvature, we are able to calculate the number of d -regular graphs G on n vertices that satisfy $Ric(G) > 0$ for $n < 15$. Due to the rapid growth in the number of regular graphs, it becomes computationally infeasible to determine the exact number of regular graphs with positive Lin–Lu–Yau curvature for larger values of n . Instead, we approximate this by randomly generating one million d -regular graphs on n vertices and calculating their Lin–Lu–Yau curvature. For the generation of the random regular graphs, we use the algorithm proposed by Steger and Wormald in [18], generating the regular graphs approximately uniformly at random.

The results of our analysis are shown in Table 2. If only a percentage (and not an exact count) is provided, the value was approximated by random sampling.

One notable observation is that regular graphs with relatively large degree always satisfy $Ric(G) > 0$. This observation will be discussed in the following section.

Additionally, observe that for a fixed degree d , there exists a threshold $\mathcal{M}_d$, such that no d -regular graph $G = (V, E)$ satisfies $|V| > \mathcal{M}_d$ and $Ric(G) > 0$. An investigation of this is conducted in Sect. 6.

5 Condition for positive Lin–Lu–Yau curvature

In this section, we present our main result: a condition on the degree of a regular graph G , ensuring that G satisfies $Ric(G) > 0$. We then discuss the sharpness of this lower bound.

Theorem 1.2 *Let $G = (V, E)$ be a finite graph. If G is d -regular with*

$$d > \frac{2|V|}{3} - 2,$$

then $Ric(G) > 0$.

To prove this, we need the following lemmata.

Lemma 5.1 *Let $G = (V, E)$ be a finite graph. If G is d -regular with*

¹ <https://github.com/moritzmath/Regular-Graph-Curvature>.

Table 2 Number of connected d -regular graphs on n vertices, up to isomorphism, that satisfy $\text{Ric}(G) > 0$, followed by the percentage of the total number of d -regular graphs on n vertices in parentheses

d	3	4	5	6	7	8
n						
4	1 (100%)	–	–	–	–	–
5	–	1 (100%)	–	–	–	–
6	2 (100%)	1 (100%)	1 (100%)	–	–	–
7	–	2 (100%)	–	1 (100%)	–	–
8	2 (40%)	6 (100%)	3 (100%)	1 (100%)	1 (100%)	–
9	–	15 (93.8%)	–	4 (100%)	–	1 (100%)
10	1 (5.3%)	32 (54%)	60 (100%)	21 (100%)	5 (100%)	1 (100%)
11	–	68 (25.7%)	–	266 (100%)	–	6 (100%)
12	0 (0%)	44 (2.9%)	6569 (83.7%)	7,849 (100%)	1,547 (100%)	94 (100%)
13	–	23 (0.2%)	–	366,804 (99.7%)	–	10,786 (100%)
14	0 (0%)	9 (0.01%)	719,285 (20.8%)	20,998,083 (97.2%)	21,609,251 (99.9%)	3,459,385 (100%)
15	–	12 (0.0015%)	–	≈ 86.5%	–	≈ 100%
16	0 (0%)	4 (0.00005%)	≈ 0.23 %	≈ 62.6 %	≈ 99.7 %	≈ 99.9 %
17	–	3 (0.000004%)	–	≈ 31.8 %	–	≈ 99.9 %

If no exact count is provided, the values are estimated by random sampling; a dash indicates that no d -regular graph on n vertices exists

$$d \geq \frac{|V| - 1}{2},$$

then $\text{diam}(G) \leq 2$.

Proof Assume that $\text{diam}(G) > 2$. Hence, there exist vertices $x, y \in V$ such that $d(x, y) > 2$, and therefore $S_1(x) \cap S_1(y) = \emptyset$. As $x, y \notin S_1(x) \cup S_1(y)$, we have

$$|S_1(x) \cup S_1(y)| \leq |V| - 2.$$

This contradicts

$$|S_1(x) \cup S_1(y)| = 2d \geq |V| - 1.$$

Therefore, our assumption was incorrect, and it follows that $\text{diam}(G) \leq 2$ must hold. $\square$

Lemma 5.2 *Let $G = (V, E)$ be a finite graph. Assume that G is d -regular with*

$$d > \frac{2|V|}{3} - 2.$$

If an edge $x \sim y$ satisfies $|\Delta(x, y)| > 2d - |V|$, then $\kappa(x, y) > 0$.

Proof Using the integrality of d together with Lemma 5.1, we conclude that $\text{diam}(G) \leq 2$. If $|\Delta(x, y)| \geq 2d - |V| + 2$, we obtain

$$\begin{aligned} \kappa(x, y) &\geq \frac{1}{d} \left(d + 1 - 2|R_x(x, y)| \right) \\ &= \frac{1}{d} \left(d + 1 - 2(d - 1 - |\Delta(x, y)|) \right) \\ &\geq \frac{1}{d} \left(3d + 7 - 2|V| \right). \end{aligned}$$

Utilizing the lower bound on d , we conclude that $\kappa(x, y) > 0$.

Now, assume $|\Delta(x, y)| = 2d - |V| + 1$ and $\kappa(x, y) \leq 0$. As $\text{diam}(G) \leq 2$, we have

$$\kappa(x, y) \geq \frac{1}{d} \left(d + 1 - 2|R_x(x, y)| \right),$$

and therefore we obtain

$$d + 1 \leq 2|R_x(x, y)| = |R_x(x, y)| + |R_y(x, y)|. \quad (5.1)$$

Let $j \in \Delta(x, y)$ be arbitrary. According to inequality 5.1, j cannot be adjacent to every vertex in $R_x(x, y) \cup R_y(x, y)$. Assume, without loss of generality, that there

exists an $i \in R_x(x, y)$ such that $i \approx j$. Thus, i has at most $d - 3$ neighbors in $S_1(x)$, as it is not adjacent to y , j and itself. Now, observe that

$$|V \setminus (S_1(x) \cup S_1(y))| = |V| - 2d + |\Delta(x, y)| = 1$$

by assumption. Therefore, there exist a $k \in R_y(x, y)$ such that $i \sim k$. Hence, an optimal assignment $\phi \in \mathcal{O}_{xy}$ must satisfy $|\{z \in R_x(x, y) : d(z, \phi(z)) = 1\}| \geq 1$, leading to

$$\begin{aligned} \kappa(x, y) &= \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right) \\ &\geq \frac{1}{d} \left(d + 1 - 2(|R_x(x, y)| - 1) - 1 \right) \\ &= \frac{1}{d} \left(3d + 6 - 2|V| \right). \end{aligned}$$

Using the lower bound on d , we conclude that $\kappa(x, y) > 0$, contradicting our assumption. $\square$

Lemma 5.3 *Let $G = (V, E)$ be a finite graph. Assume that G is d -regular with*

$$d > \frac{2|V|}{3} - 2.$$

If an edge $x \sim y$ satisfies $|\Delta(x, y)| = 2d - |V|$, then $\kappa(x, y) > 0$.

Proof Using the integrality of d together with Lemma 5.1, we conclude that $\text{diam}(G) \leq 2$. If $R_x(x, y) = \emptyset$, then $\kappa(x, y) = \frac{d+1}{d} > 0$. Thus, we now assume $R_x(x, y) \neq \emptyset$.

Let $i \in R_x(x, y)$ be arbitrary. Since $i \approx y$, it is adjacent to at most $d - 2$ vertices in $S_1(x)$. By assumption, we have $V = S_1(x) \cup S_1(y)$, so there must exist a vertex $j \in R_y(x, y)$ such that $i \sim j$. Similarly, every vertex in $R_y(x, y)$ is adjacent to a vertex in $R_x(x, y)$.

Now, denote by H the bipartite subgraph with vertex set $R_x(x, y) \cup R_y(x, y)$ and containing all edges connecting a vertex in $R_x(x, y)$ with a vertex in $R_y(x, y)$. According to the preceding argument, every vertex in H is of degree greater than or equal to one in H . Denote by $\Delta(H)$ the maximal degree of H . We proceed with a proof by cases.

Case 1: $\Delta(H) = 1$. The induced bipartite subgraph is 1-regular. According to Corollary 2.3, there exists a perfect matching of $R_x(x, y)$ into $R_y(x, y)$. Hence, we obtain

$$\kappa(x, y) = \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right)$$

$$= \frac{1}{d} \left(d + 1 - |R_x(x, y)| \right) > 0.$$

Case 2: $1 < \Delta(H) \leq |R_x(x, y)| - 2$. Let $z_1 \in R_x(x, y)$ be, without loss of generality, a vertex of degree $\Delta(H)$ in H . There exist distinct vertices $i, j \in R_y(x, y)$ such that z_1 is not adjacent to both i and j . Choose $z_2 \in R_x(x, y)$ such that $z_2 \sim i$. If the degree of z_2 in H is greater than one, choose $z_3 \in R_x(x, y) \setminus \{z_1, z_2\}$ arbitrary. otherwise, choose $z_3 \in R_x(x, y)$ such that $z_3 \sim j$. By construction, every $X \subseteq \{z_1, z_2, z_3\}$ satisfies $|N(X)| \geq |X|$ in H . Hall's marriage theorem implies that an optimal assignment $\phi \in \mathcal{O}_{xy}$ must satisfy $|\{z \in R_x(x, y) : d(z, \phi(z)) = 1\}| \geq 3$, leading to

$$\begin{aligned} \kappa(x, y) &= \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right) \\ &\geq \frac{1}{d} \left(d + 1 - 2(|R_x(x, y)| - 3) - 3 \right) \\ &= \frac{1}{d} (3d + 6 - 2|V|). \end{aligned}$$

Using the lower bound on d , we conclude that $\kappa(x, y) > 0$.

Case 3: $\Delta(H) \geq |R_y(x, y)| - 1$. Let $z_1 \in R_x(x, y)$ be, without loss of generality, a vertex of degree $\Delta(H)$ in H . Recall that $\text{diam}(G) \leq 2$ and that z_1 can be matched with a vertex at distance one in $R_y(x, y)$, since the degree of z_1 is at least one in H . Therefore, we obtain

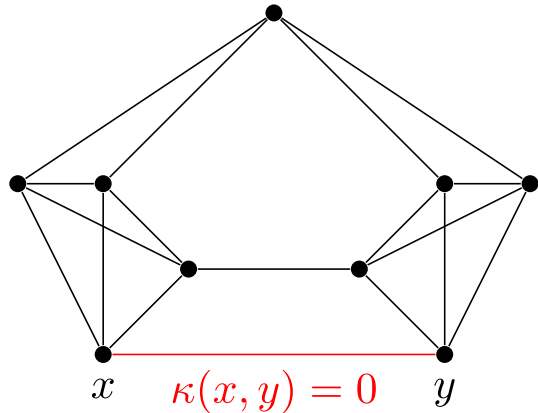
$$\kappa(x, y) \geq \frac{1}{d} (d + 2 - 2|R_x(x, y)|).$$

If $|R_x(x, y)| + |R_y(x, y)| \leq d + 1$, we conclude that $\kappa(x, y) > 0$. Thus, we now assume

$$d < |R_x(x, y)| + |R_y(x, y)| - 1. \quad (5.2)$$

Note that this implies $|R_x(x, y)| \geq 3$, since $d \geq |R_x(x, y)| + 1$. Since z_1 is adjacent to x and to at least $|R_y(x, y)| - 1$ vertices in $R_y(x, y)$, it cannot be adjacent to every vertex in $R_x(x, y)$ according to inequality 5.2. Thus, we can choose $z_2 \in R_x(x, y)$ such that $z_1 \not\sim z_2$ in G . Hence, z_2 is adjacent to at most $d - 3$ vertices in $S_1(x)$. Recall that, by assumption, $V = S_1(x) \cup S_1(y)$, implying that the degree of z_2 in H is greater than or equal to two. Choose $z_3 \in R_x(x, y) \setminus \{z_1, z_2\}$ arbitrary. Then, by construction, every $X \subseteq \{z_1, z_2, z_3\}$ satisfies $|N(X)| \geq |X|$ in H . Hall's marriage theorem implies that an optimal assignment $\phi \in \mathcal{O}_{xy}$ between $R_x(x, y)$ and $R_y(x, y)$ satisfies $|\{z \in R_x(x, y) : d(z, \phi(z)) = 1\}| \geq 3$, leading to $\kappa(x, y) > 0$, as in *Case 2*. $\square$

Fig. 2 Illustration of a 4-regular graph G on 9 vertices that does not satisfy $\text{Ric}(G) > 0$



Proof of Theorem 1.2 Since $S_1(x) \cup S_1(y) \subseteq V$, the inequality

$$2d - |V| \leq |\Delta(x, y)|$$

always holds true. If $2d - |V| < |\Delta(x, y)|$, the claim follows from Lemma 5.2. If $2d - |V| = |\Delta(x, y)|$, the claim follows from Lemma 5.3. This concludes the proof of Theorem 1.2. $\square$

Remark 5.4 For $d = \frac{2|V|}{3} - 2$, it is possible for edges $x \sim y$ with $\kappa(x, y) \leq 0$ to occur. An example is illustrated in Fig. 2.

We can generalize Theorem 1.2 to the 0-Ollivier–Ricci-curvature, using the following result by Cushing et al.

Theorem 5.5 [19, Theorem 1.1]. *Let $G = (V, E)$ be a regular graph. Then we have the following implication for all edges $x \sim y$:*

$$\kappa(x, y) > 0 \implies \kappa_0(x, y) \geq 0.$$

Hence, we obtain the following result for the Ollivier–Ricci curvature.

Theorem 1.3 *Let $G = (V, E)$ be a finite graph. If G is d -regular with*

$$d > \frac{2|V|}{3} - 2,$$

then $\kappa_0(x, y) \geq 0$ for every edge $x \sim y$.

Next, we discuss the asymptotic sharpness of the lower bound.

Theorem 1.4 *Let $n \in \mathbb{N}$ such that*

$$d = \frac{2n - 8}{3} \in 2\mathbb{N}.$$

and $d \geq 12$. Then there exists a d -regular graph on n vertices such that $\kappa(x, y) = 0$ and $\kappa_0(x, y) < 0$ for an edge $x \sim y$.

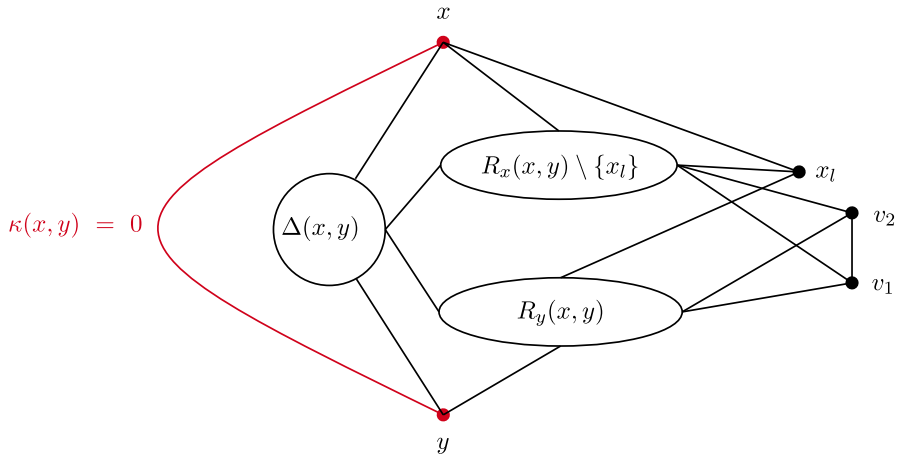


Fig. 3 Illustration of the graph constructed in Theorem 1.4

Proof Let the assumptions of Theorem 1.4 hold. We construct a d -regular graph on n vertices that contains an edge $x \sim y$ such that $\kappa(x, y) = 0$. To this end, define the vertex set

$$V = \{x, y, z_0, \dots, z_{l-3}, x_0, \dots, x_l, y_0, \dots, y_l, v_1, v_2\},$$

where $l = \frac{d}{2}$. Next, we add the edges

- $x \sim y$,
- $x \sim z_i$ for $i = 0, \dots, l-3$,
- $x \sim x_i$ for $i = 0, \dots, l$,
- $y \sim z_i$ for $i = 0, \dots, l-3$,
- $y \sim y_i$ for $i = 0, \dots, l$.

to the set of edges E . Therefore, x and y are of degree d , and

$$\Delta(x, y) = \{z_0, \dots, z_{l-3}\}, \quad R_x(x, y) = \{x_0, \dots, x_l\}, \quad R_y(x, y) = \{y_0, \dots, y_l\}.$$

Next, we add for every $z_i \in \Delta(x, y)$ the edges $z_i \sim x_j$ for $j = 0, \dots, l-1$ if $j \neq i$ as well as the edges $z_i \sim y_j$ for $j = 0, \dots, l$ if

$$j \notin \{2i \pmod{l+1}, 2i+1 \pmod{l+1}\}.$$

Thus, every $z_j \in \Delta(x, y)$ is of degree d , as it has $d-2$ neighbors in $R_x(x, y) \cup R_y(x, y)$ and is adjacent to both x and y .

On the other hand, every x_i is adjacent to $|\Delta(x, y)| - 1$ vertices in $\Delta(x, y)$ if $i \leq l-3$, and to all vertices in $\Delta(x, y)$ if $i \in \{l-2, l-1\}$. Every y_i is adjacent to $|\Delta(x, y)| - 2$ vertices in $\Delta(x, y)$ if $i \leq l-6$ and to $|\Delta(x, y)| - 1$ otherwise.

Next, we add the edges $x_i \sim x_j$ for $i \neq j$, except $x_{l-2} \sim x_{l-1}$. Analogously, we add the edges $y_i \sim y_j$ for $i \neq j$, as well as $x_l \sim y_j$ for $j = 0, \dots, l-2$.

Thus, every $x_i \in R_x(x, y) \setminus \{x_l\}$ has degree $d-2$ and x_l has degree d . On the other hand, y_i has degree $d-1$ if $i \in \{l-5, l-4, l-3, l-2\}$ and $d-2$ otherwise.

Finally, we add the edges

- $v_1 \sim x_i$ and $v_2 \sim x_i$ for $i = 0, \dots, l-1$.
- $v_1 \sim y_i$ and $v_2 \sim y_i$ for $i = 0, \dots, l-6$ and $i = l-1, l$.
- $v_1 \sim y_{l-5}$, $v_1 \sim y_{l-4}$, $v_2 \sim y_{l-3}$, $v_2 \sim y_{l-2}$ and $v_1 \sim v_2$.

The constructed graph $G = (V, E)$ is d -regular. Now, denote by H the bipartite subgraph with vertex set $R_x(x, y) \cup R_y(x, y)$ and containing all edges connecting a vertex in $R_x(x, y)$ with a vertex in $R_y(x, y)$. By construction, every x_j for $0 \leq j \leq l-1$ is of degree zero in H . Therefore, according to Hall's marriage theorem, an optimal assignment $\phi \in \mathcal{O}_{xy}$ must satisfy $|\{z \in R_x(x, y) : d(z, \phi(z)) = 1\}| = 1$. Thus, we obtain

$$\begin{aligned} \kappa(x, y) &= \frac{1}{d} \left(d + 1 - \inf_{\phi \in \mathcal{A}_{xy}} \sum_{z \in R_x(x, y)} d(z, \phi(z)) \right) \\ &= \frac{1}{d} \left(d + 1 - 2(|R_x(x, y)| - 1) - 1 \right) \\ &= 0. \end{aligned}$$

Furthermore, every vertex in $R_x(x, y)$ can be assigned to x at distance one and y can be assigned to every vertex in $R_y(x, y)$ at distance one. Therefore, we obtain

$$\begin{aligned} \kappa_0(x, y) &= \frac{1}{d} \left(d - \inf_{\phi} \sum_{z \in S_1(x) \setminus S_1(y)} d(z, \phi(z)) \right) \\ &= \frac{1}{d} \left(d - 2(|S_1(x) \setminus S_1(y)| - 3) - 3 \right) \\ &= -\frac{1}{d}. \end{aligned}$$

□

6 Maximal regular graphs with positive Ricci curvature

We finish this article with an investigation of the maximal number of vertices that an d -regular graph G satisfying $\text{Ric}(G) > 0$ can have,

$$\mathcal{M}_d = \max_{\substack{G \text{ } d\text{-regular,} \\ \text{Ric}(G) > 0}} |V(G)|.$$

The following Proposition shows that $\mathcal{M}_d$ is always finite.

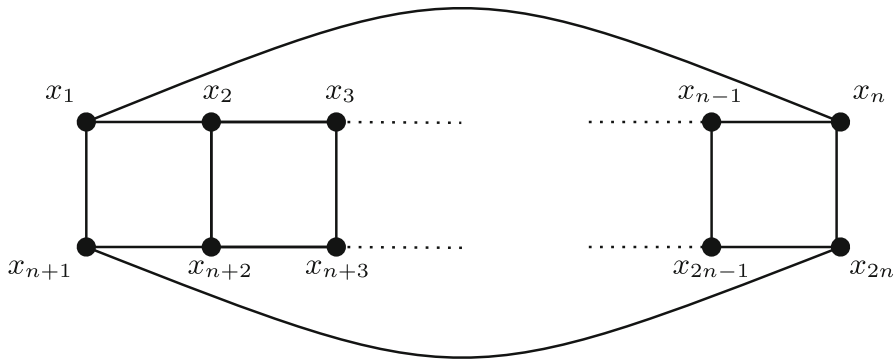


Fig. 4 The prism graph Y_n

Proposition 6.1 *Let $d \in \mathbb{N}$. Then*

$$\mathcal{M}_d \leq 1 + d \sum_{i=0}^{2d-1} (d-1)^i.$$

For the proof of Proposition 6.1, we make use of the following Bonnet–Myers type theorem on graphs by Lin, Lu, and Yau.

Theorem 6.2 (Discrete Bonnet–Myers Theorem [7]). *Let $G = (V, E)$ be a locally finite graph. If $\text{Ric}(G) \geq k > 0$, then the diameter of graph G is bounded as follows:*

$$\text{diam}(G) \leq \frac{2}{k}.$$

Proof of Proposition 6.1 Let G be an arbitrary d -regular graph, satisfying $\text{Ric}(G) > 0$. According to Corollary 3.6, we have $\kappa(x, y) \in \mathbb{Z}/d$ for every edge $x \sim y$. Therefore, $\text{Ric}(G) \geq \frac{1}{d}$ must hold. Using Theorem 6.2, we obtain

$$\text{diam}(G) \leq 2d.$$

The proposition is now an immediate consequence of the Moore bound. $\square$

The following theorem quantifies $\mathcal{M}_d$ for d equal to two and three.

Theorem 1.5 *Let $G = (V, E)$ be a d -regular graph, satisfying $\text{Ric}(G) > 0$. If $d = 2$, then $|V| \leq 5$. If $d = 3$, then $|V| \leq 10$. Consequently, we have $\mathcal{M}_2 = 5$ and $\mathcal{M}_3 = 10$.*

For the proof of Theorem 1.5, we need the following classification result by Cushing et al. [20].

Theorem 6.3 *Let $G = (V, E)$ be a 3-regular graph. Then $\kappa_0(x, y) \geq 0$ for all edges $x \sim y$ if and only if G is a prism graph Y_n for some $n \geq 3$ or a Möbius ladder M_n for some $n \geq 2$.*

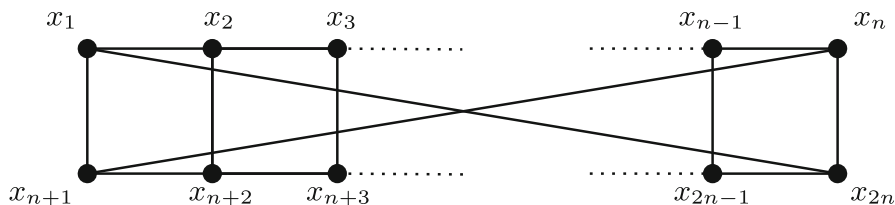


Fig. 5 The Möbius Ladder M_n

Figure 4 illustrates the prism graph Y_n , while Fig. 5 provides a visualization of the Möbius ladder M_n . With this classification result established, we are now prepared to prove Theorem 1.5.

Proof of Theorem 1.5 The only 2-regular, connected graphs are the cycle graphs C_n . It is straightforward to verify that $Ric(C_n) = 0$ for $n \geq 6$. Hence, if $G = (V, E)$ is 2-regular satisfying $Ric(G) > 0$, then $G = C_n$ for some $2 \leq n \leq 5$ and therefore $|V| \leq 5$ holds true.

We now turn to the case where $d = 3$. Let $x \sim y$ be an arbitrary edge in G . By assumption, we have $\kappa(x, y) > 0$. Since G is regular, it follows from Theorem 5.5 that $\kappa_0(x, y) \geq 0$. Thus, according to Theorem 6.3, G is a prism graph Y_n for some $n \geq 3$ or a Möbius ladder M_n for some $n \geq 2$. The prism graph Y_n does not satisfy $Ric(Y_n) > 0$ for $n \geq 6$ and the Möbius ladder M_n does not satisfy $Ric(M_n) > 0$ for $n \geq 5$. Therefore, G has at most 10 vertices. $\square$

The values $\mathcal{M}_d$ for $d \geq 4$ are still unknown. The Cartesian product of n 5-cycles $(C_5)^{\square n}$ is $2n$ -regular, has 5^n vertices and satisfies

$$Ric((C_5)^{\square n}) = \frac{1}{2n} > 0.$$

This is an immediate consequence of $Ric(C_5) = \frac{1}{2}$ and Corollary 3.2 in [7]. Thus, $\mathcal{M}_d$ exhibits exponential growth, satisfying

$$\mathcal{M}_d \geq \sqrt{5}^d$$

for d even.

Recall that $\kappa(x, y) \in \mathbb{Z}/d$ for every edge $x \sim y$ of an d -regular graph. Hence $(C_5)^{\square n}$ exhibits constant, minimal positive Lin–Lu–Yau curvature for an $2n$ -regular graph. This led us to formulate the following conjecture.

Conjecture 6.4 For $d \in 2\mathbb{N}$,

$$\mathcal{M}_d = \sqrt{5}^d.$$

Moreover, for odd d , $\mathcal{M}_d$ also exhibits exponential growth, as e.g. the n -dimensional hypercube $\mathcal{Q}_n$ for odd n shows.

Appendix

A Comparative analysis of implementations

In this section, we present a comparative analysis of our implementation for computing the Lin–Lu–Yau curvature against existing implementations. We compare our program with those by Ni et al., presented in [17], and by Cushing et al., presented in [20]. For this comparison, we generate random d -regular graphs on n vertices, employing the algorithm proposed by Steger and Wormald in [18]. We then calculate the Lin–Lu–Yau curvature of these graphs using the different implementations. To ensure robustness, computation times are averaged over 100 graphs for each combination of parameters n and d . The computations are performed on an Apple M2 chip (8-core CPU at up to 3.5 GHz) with 16 GB RAM. Due to the limited efficiency of the implementation by Cushing et al., we exclude it from the analysis on larger graphs, as the computation time is prohibitively high. For smaller graphs ($n \leq 20$) our implementation achieves a speedup of 20–50 times over the other implementations. The results of our comparative analysis are presented in Fig. 6.

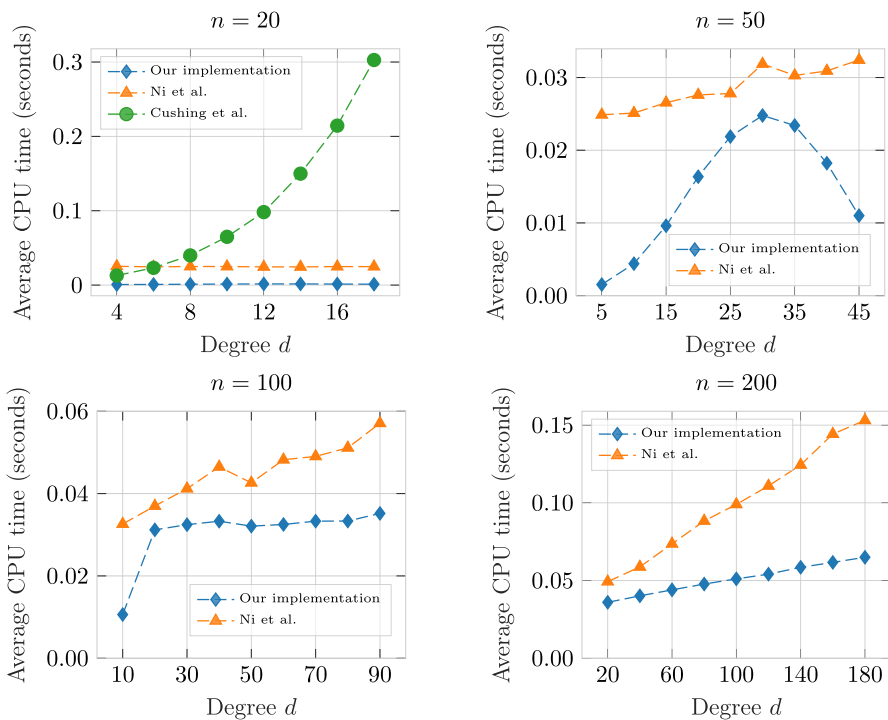


Fig. 6 A comparative analysis of the different implementations for computing the Lin–Lu–Yau curvature on regular graphs. We display the average CPU time of each implementation for computing the Lin–Lu–Yau curvature of an d -regular graph on n vertices

Acknowledgements I would like to express my deepest gratitude to Prof. Renesse and Dr. Münch for their invaluable support and insightful feedback throughout the course of this work.

Funding Open Access funding enabled and organized by Projekt DEAL. Partial financial support was received from the BMBF (Federal Ministry of Education and Research) in DAAD project 57616814 (SECAI, School of Embedded Composite AI).

Data Availability The source code for the implementation presented in Sect. 4 is publicly available at <https://github.com/moritzmath/Regular-Graph-Curvature>.

Declarations

Conflict of interest The author certifies that he has no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. Bakry, D., Émery, M.: Diffusions hypercontractives. In: *Séminaire de Probabilités XIX 1983/84*, pp. 177–206 (1985). <https://doi.org/10.1007/BFb0075847>
2. Erbar, M., Maas, J.: Ricci curvature of finite Markov chains via convexity of the entropy. In: *Archive for Rational Mechanics and Analysis*, vol. 206, pp. 997–1038 (2012). <https://doi.org/10.1007/s00205-012-0554-z>
3. Mielke, A.: Geodesic convexity of the relative entropy in reversible Markov chains. In: *Calculus of Variations and Partial Differential Equations*, vol. 48, pp. 1–31 (2013). <https://doi.org/10.1007/s00526-012-0538-8>
4. Forman, R.: Bochner's method for cell complexes and combinatorial Ricci curvature. *Discrete Comput. Geom.* **29**, 323–374 (2003). <https://doi.org/10.1007/s00454-002-0743-x>
5. von Renesse, M.-K., Sturm, K.-T.: Transport inequalities, gradient estimates, entropy and Ricci curvature. *Commun. Pure Appl. Math.* **58**(7), 923–940 (2005). <https://doi.org/10.1002/cpa.20060>
6. Ollivier, Y.: Ricci curvature of Markov chains on metric spaces. *J. Funct. Anal.* **256**(3), 810–864 (2009)
7. Lin, Y., Linyuan, L., Yau, S.-T.: Ricci curvature of graphs. *Tohoku Math. J.* **63**(4), 605–627 (2011). <https://doi.org/10.2748/tmj/1325886283>
8. Bonini, V., et al.: Condensed Ricci curvature of complete and strongly regular graphs. *Involve J. Math.* **13**(4), 559–576 (2020). <https://doi.org/10.2140/involve.2020.13.559>
9. Münch, F., Wojciechowski, R.K.: Ollivier Ricci curvature for general graph Laplacians: heat equation, Laplacian comparison, non-explosion and diameter bounds. *Adv. Math.* **356**, 106759 (2019). <https://doi.org/10.1016/j.aim.2019.106759>
10. Hall, P.: On Representatives of Subsets. *J. Lond. Math. Soc.* **10**(1), 26–30 (1935). <https://doi.org/10.1112/jlms/s1-10.37.26>
11. Vilani, C.: *Topics in Optimal Transportation*, Graduate Studies in Mathematics. American Mathematical Society, Providence (2003)
12. Bourne, D.P., et al.: Ollivier–Ricci idleness functions of graphs. *SIAM J. Discrete Math.* **32**(2), 1408–1424 (2018). <https://doi.org/10.1137/17M1134469>
13. Meringer, M.: Fast generation of regular graphs and construction of cages. *J. Graph Theory* **30**, 137–146 (1999)

14. McKay, B.D., Wormald, N.C.: Asymptotic enumeration by degree sequence of graphs of high degree. *Eur. J. Comb.* **11**(6), 565–580 (1990). [https://doi.org/10.1016/S0195-6698\(13\)80042-X](https://doi.org/10.1016/S0195-6698(13)80042-X)
15. Liebenau, A., Wormald, N.C.: Asymptotic enumeration of graphs by degree sequence, and the degree sequence of a random graph. *J. Eur. Math. Soc.* **26**, 1–40 (2024)
16. McKay, B.D., Wormald, N.C.: Automorphisms of random graphs with specified vertices. *Combinatorica* **4**, 325–338 (1984). <https://doi.org/10.1007/BF02579144>
17. Ni, C.-C., et al.: Community detection on networks with Ricci flow. *Sci. Rep.* **9**(1), 1–12 (2019). <https://doi.org/10.1038/s41598-019-46380-9>
18. Steger, A., Wormald, N.C.: Generating random regular graphs quickly. *Comb. Probab. Comput.* **8**(4), 377–396 (1999). <https://doi.org/10.1017/S0963548399003867>
19. Cushing, D., et al.: Curvatures, graph products and Ricci flatness. *J. Graph Theory* **96**(4), 522–553 (2021). <https://doi.org/10.1002/jgt.22630>
20. Cushing, D., et al.: The graph curvature calculator and the curvatures of cubic graphs. *Exp. Math.* **31**(2), 583–595 (2022). <https://doi.org/10.1080/10586458.2019.1660740>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.